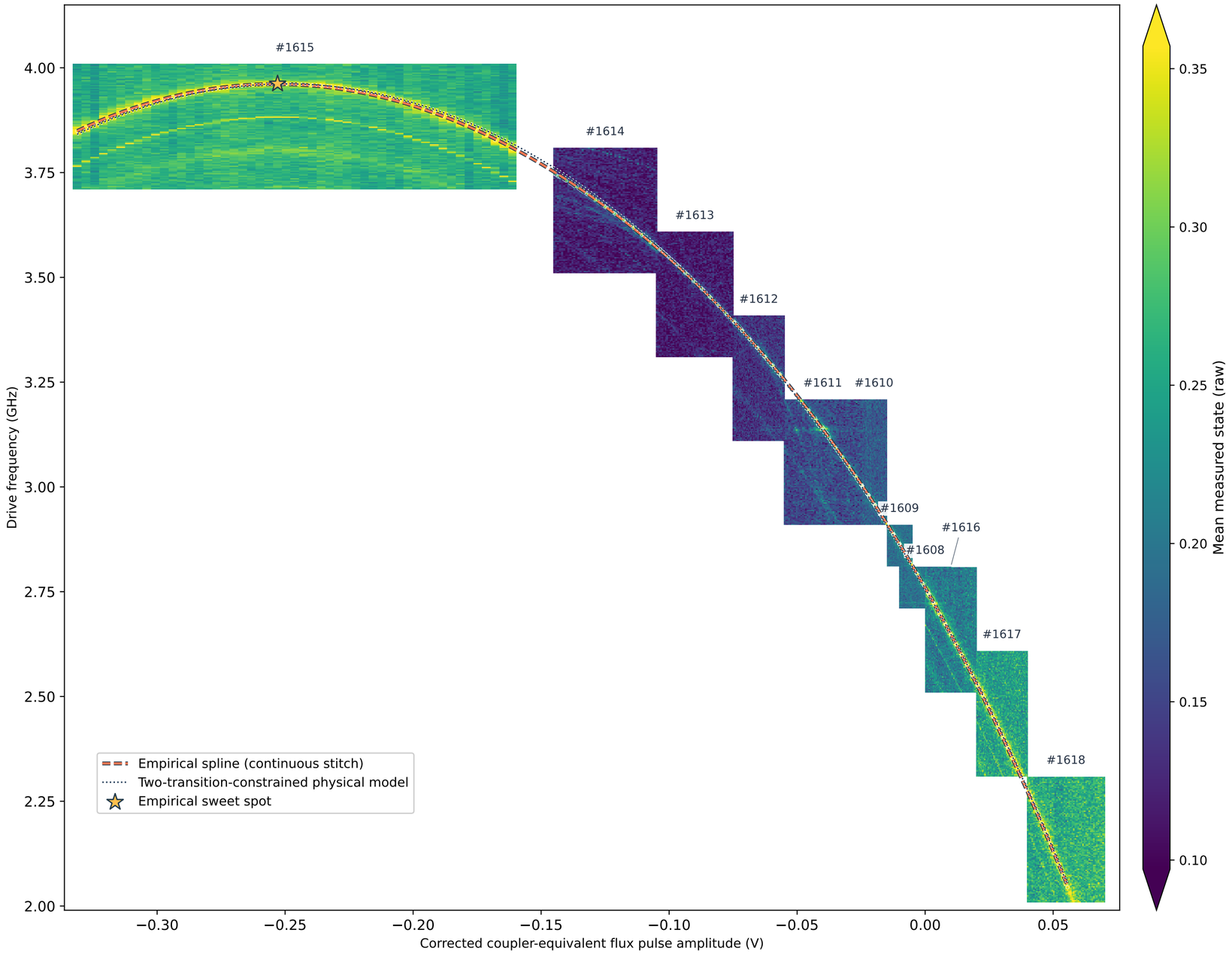
Coupler flux spectroscopy | all fitting functions and transition constraint

coupler_q4_q5 | 2026-11-04 | #1608-#1618



註：#1615 因 q4、q5 各加 +0.3 V 的 flux crosstalk，Flux 軸左移 16.21 mV；合計等效係數約 -5.4%。

橘色虛線為經驗拼接曲線；深色點線為加入 f01、f02 約束的物理模型。原始頻率與訊號保留。

白色區域未量測，跨越接縫的曲線為插值；物理模型仍有系統性殘差，d 尚未可靠辨識。

Fitting functions and constraints

1. Resonance extraction at each flux

$$P(f) = b_0 + b_1(f - f_s) + \frac{A}{1 + [(f - f_r)/\gamma]^2}$$

$f_r(V)$ is the fitted line center; FWHM $= 2\gamma$.

2. Flux-crosstalk registration

$$V_c = V_{raw} - 0.01621188\, I_{1615} \quad [\mathrm{V}]$$

$\kappa_{q4} + \kappa_{q5} = \Delta V / 0.30 = -5.40\%$

q4 and q5 each receive +0.30 V; frequencies retained.

3. Local join: #1614 and #1615

$$f_{join}/\mathrm{GHz} = 3.9618717 - 0.4194001 z^2 + 0.0217719 z^4$$

$$z = (V_c + 0.25301879)/0.15$$

4. Full empirical curve: cubic B-spline

$$f_{spline}(V_c) = \sum_j c_j B_{j,3}(V_c; t)$$

$t = [a, a, a, a, -0.1402, -0.096, -0.0322, b, b, b, b]$

$a = -0.33121188, \; b = 0.05560000$

c [GHz] = [3.849187, 4.034448, 3.974462, 3.505628, 3.053948, 2.488679, 2.049398]

5. Physical SQUID / transmon model

$$H/h = 4e_C(\hat{n} - n_g)^2 - e_{J,\,eff}(V_c)\cos\hat{\varphi}$$

$$e_{J,\,eff} = e_{J\Sigma}\sqrt{\cos^2\theta + d^2\sin^2\theta}$$

$\theta = \pi(V_c - V_s)/V_{\Phi_0}, \quad d = |E_{J1} - E_{J2}|/E_{J\Sigma}$

$f_{01} = \lambda_1 - \lambda_0, \quad f_{02}/2 = (\lambda_2 - \lambda_0)/2$

λ_i: eigenvalues of H/h; $e_C = E_C/h$, $e_{J\Sigma} = E_{J\Sigma}/h$.

Large EJ/EC approximation (not the numerical fit):

$$f_{01} \simeq \sqrt{8 e_C e_{J\Sigma}}(\cos^2\theta + d^2\sin^2\theta)^{1/4} - e_C$$

6. Provided two-transition constraint

$f_{01} = 2.760584, \quad f_{02}/2 = 2.672584$ GHz

$\alpha = f_{02} - 2f_{01} = -176$ MHz

$e_C \approx -\alpha = 0.176$ GHz (leading approximation)

$e_C = 0.152256, \quad e_{J\Sigma} = 13.9370$ GHz

$V_{\Phi_0} = 0.751305$ V, $\quad V_s = -0.249839$ V

$n_g = 0.500, \quad d \to 0$ (boundary; not resolved)

Full-spectrum RMSE = 4.25 MHz; systematic residuals remain.

Energy values are conditional on the isolated-transmon model.

Numerical EC uses both supplied transitions, not EC = -alpha.

Displayed coefficients rounded; full precision saved in JSON.